

Agentic AI for Economic Research

Concepts, Tools, Workflows, and Case Studies

Mini-course for PhD students and faculty

1 July 2026

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Course abstract

This four-hour minicourse introduces doctoral students and faculty members to the practical and critical use of agentic artificial intelligence in economics research. The course is designed as a research workflow map rather than a technical manual: its goal is to help participants understand what can be delegated to AI agents, what must remain under human control, and how to design AI-assisted research processes that remain transparent, reproducible, and methodologically sound.

The first part develops the core vocabulary and conceptual framework. It explains the shift from chatbots that answer prompts to agents that can inspect files, call tools, use apps, work inside IDEs or command-line environments, and iterate through plan-act-verify loops. Participants will discuss context, tokens, tool use, permissions, sandboxing, autonomy levels, rules, memory, sub-agents, MCP, CLI and API workflows, as well as the changing boundary between chat apps, desktop agent apps, IDE agents, browser/research assistants, and local or command-line agents. The course also introduces knowledge bases and memory systems, including why a curated Obsidian-style Markdown knowledge base can be preferable to a pure RAG setup when provenance, auditability, and human curation matter.

The course also covers governance and reliability: pricing and token consumption, data protection, institutional constraints, reproducibility, and the risks associated with hallucinations, automation bias, AI sloppiness, weak validation, and excessive trust in polished outputs. A central theme is that agentic AI is safest and most useful when verification is designed into the workflow through plans, diffs, logs, tests, source checks, and explicit human review gates.

The second part turns to live case studies from current economics research projects. Participants will examine how agents can support literature review and source triage, data construction and codebook documentation, Stata/R/Python debugging, replication workflows, robustness checks, table creation from data, paper rewriting, slide production, grant and referee-response preparation, and teaching material design. These examples are used to understand what different project difficulties teach about setup, prompting, tool choice, memory, verification, and when to stop delegating.



Photo: public Google profile linked from Marco Le Moglie's Google Site.

Instructor

Marco Le Moglie is Associate Professor of Economics at the Catholic University of the Sacred Heart in Milan and a research fellow at CLEAN, Bocconi University. His research focuses on the economics of crime, political economy, and public economics. His current work uses applied microeconomic methods and text/data workflows to study organised crime, political selection, public finance, and institutional responses to criminal infiltration. In this minicourse, he draws on his own research workflow to discuss how agentic AI can support economics research without replacing human judgement on theory, identification, measurement, interpretation, and responsibility.